$R \to$ orden del tensor

Ej 6

tensor de rotación

tensor cartesiano $\Rightarrow$ $A'_{\gamma_1 \gamma_2 \ldots \gamma_R} = \beta_{\gamma_1 \alpha_1} \ldots \beta_{\gamma_R \alpha_R} A_{\alpha_1 \ldots \alpha_R}$

$\underline{\underline{\beta}} \cdot \underline{\underline{\beta}}^T = \underline{\underline{\beta}}^T \cdot \underline{\underline{\beta}} = \underline{\underline{I}}$

$(\underline{\underline{\beta}})_{ij} (\underline{\underline{\beta}}^T)_{jk} = \beta_{ij} \beta_{kj} = \delta_{ik}$

orden 2 $\quad$ orden 0

$A_{ij} \to A_{ii}$

$\gamma_1 = \gamma_2 \equiv j$

orden del tensor

$R-2 \to A'_{jj\gamma_3 \ldots \gamma_R} = \underbrace{\beta_{j\alpha_1} \cdot \beta_{j\alpha_2}}_{\delta_{\alpha_1 \alpha_2}} \beta_{\gamma_3 \alpha_3} \ldots \beta_{\gamma_R \alpha_R} A_{\alpha_1 \alpha_2 \ldots \alpha_R}$

dummy

libre $\gamma_3 \ldots \gamma_R$

$A'_{jj\gamma_3 \ldots \gamma_R} = \beta_{\gamma_3 \alpha_3} \ldots \beta_{\gamma_R \alpha_R} A_{\alpha_1 \alpha_1 \alpha_3 \ldots \alpha_R}$

$$\boxed{C'_{\pi_1 \ldots \pi_{R-2}} = \beta_{\pi_1 \mu_1} \ldots \beta_{\pi_R \mu_{R-2}} C_{\mu_1 \ldots \mu_{R-2}}}$$

ley de transformación para un tensor de orden $R-2$